Condensate Partition Coefficient Pipeline

Instruction Manual

Automated 3D segmentation of nuclear biomolecular condensates and measurement of the partition coefficient from confocal Z-stacks

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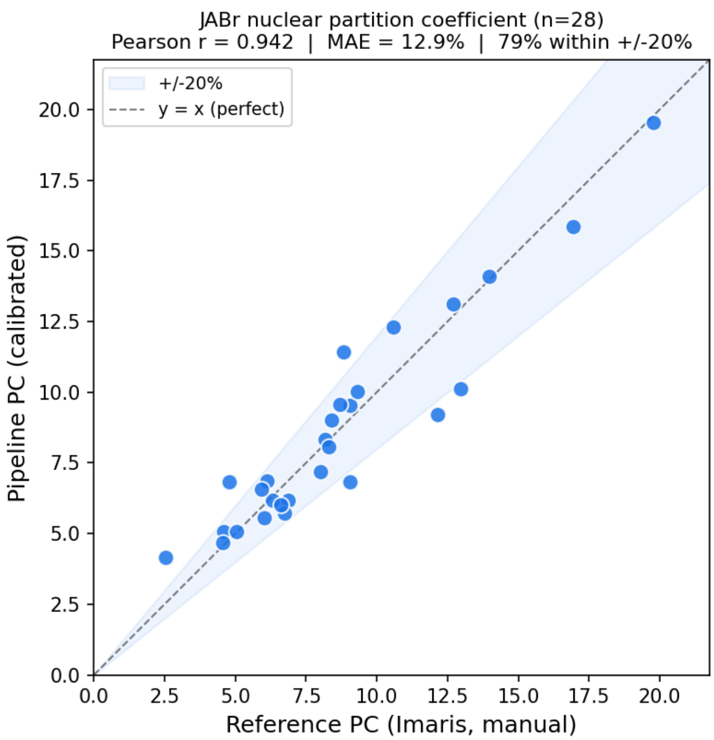
1 Overview

This pipeline replaces the manual Imaris workflow for measuring the **partition coefficient (PC)** of nuclear biomolecular condensates. Given a two-channel confocal Z-stack, it segments nuclei and condensates in 3D, computes the PC, and reports a calibrated value on the manual reference scale.

Inputs	A two-channel Z-stack: channel 0 = nuclei, channel 1 = condensate.
Outputs	Nuclear & cytoplasmic PC, 3D masks, per-object volumes, a summary figure.
Validated on	JABr: Pearson $r = 0.942$, MAE 12.9%, 79% of cells within ±20% (n = 28).
Interfaces	Graphical (`run_gui.py`) or command line (`pipeline.py`).



2 Installation (one time)

1. Install **Python 3.10 or 3.11**.

2. Install **PyTorch** for your machine from pytorch.org (choose CPU or your CUDA version). Example for CUDA 12.1:

```
pip install torch --index-url https://download.pytorch.org/whl/cu121
```

3. Install the remaining dependencies:

```
pip install -r requirements.txt
```

On first run, Cellpose downloads its model weights automatically (one-time ∼100 MB). A GPU is recommended (∼10–15 s per cell); CPU works but is slower.

3 Preparing your image files

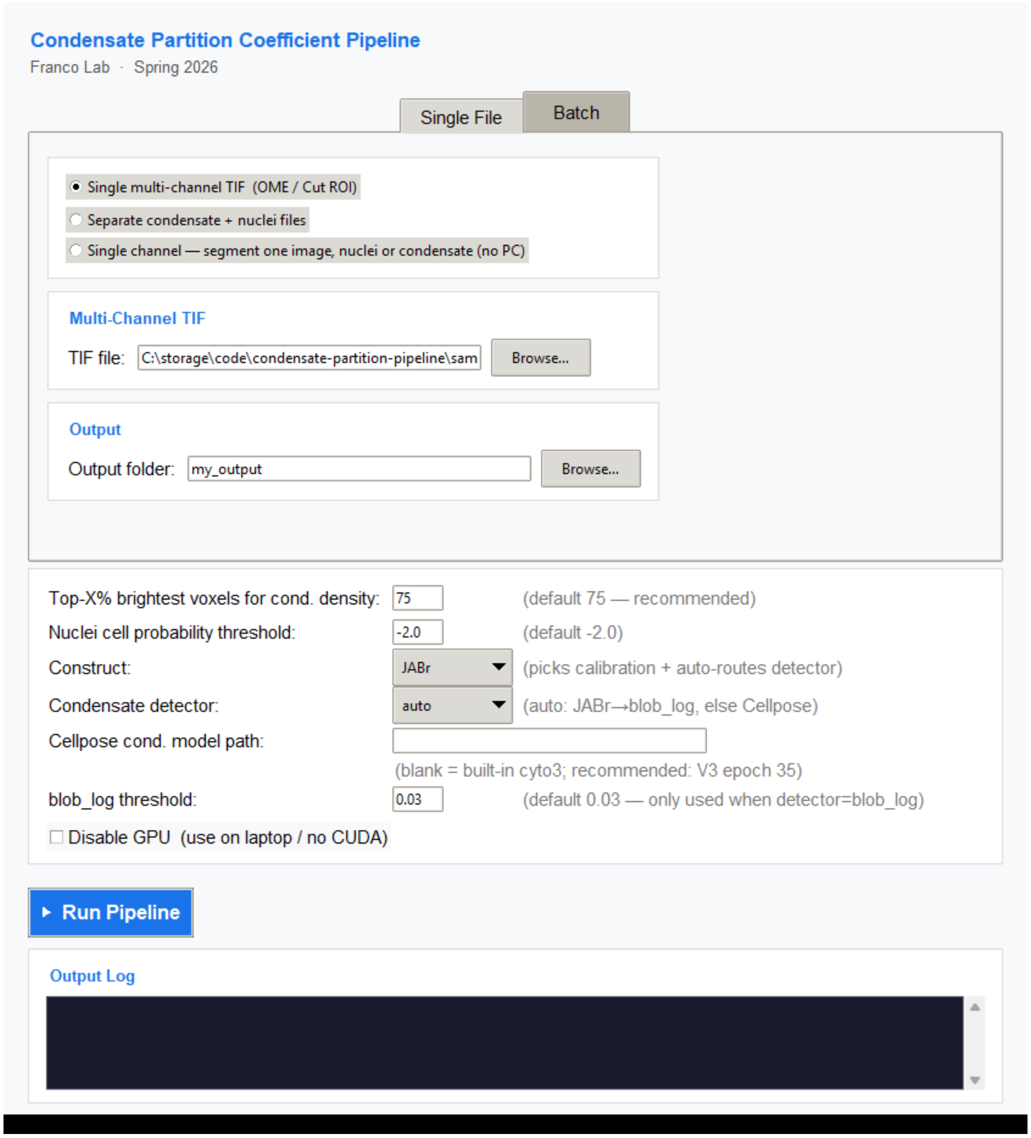
The pipeline accepts either one multi-channel TIF (both channels in one file) or two separate single-channel TIFs. **Channel order matters:** channel 0 must be nuclei, channel 1 the condensate. Most axis orderings (ZCYX, CZYX, . . .) are auto-detected. If results look wrong, swapped channels are the most common cause.

4 Running with the graphical interface

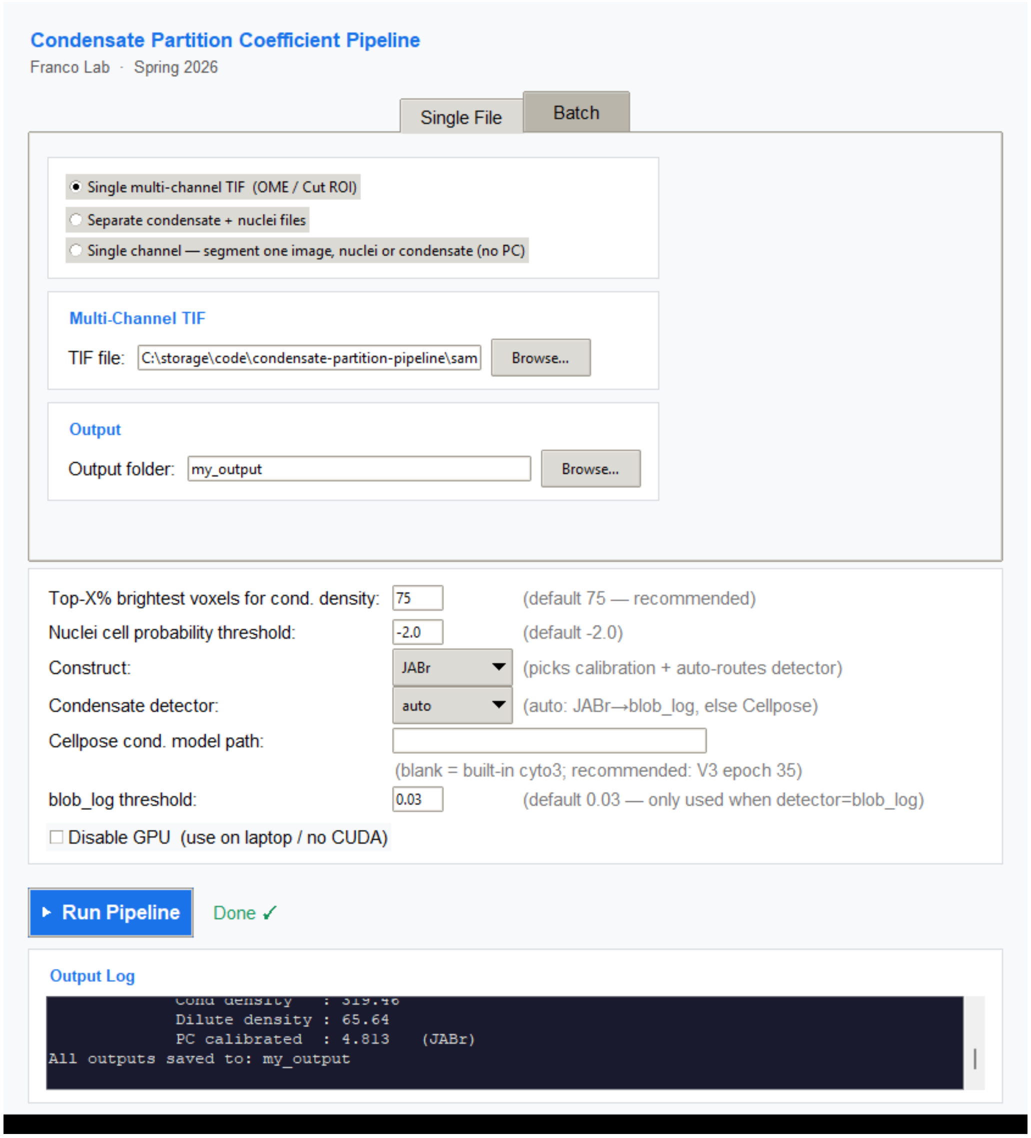
Launch with `python run_gui.py`. A window with **Single File** and **Batch** tabs opens.

4.1 Step by step (single cell)

1. On the **Single File** tab, choose the input mode (one multi-channel TIF, or separate files) and **Browse. . .** to your data.
2. Choose an **Output** folder (optional).
3. In **Settings**, set **Construct = JABr**. For JABr, leave all other settings at their defaults.
4. Click **▶ Run Pipeline**. Progress streams in the **Output Log**; the status reads **Done** when finished.
5. Open your output folder for the masks, tables, and `results.png`.



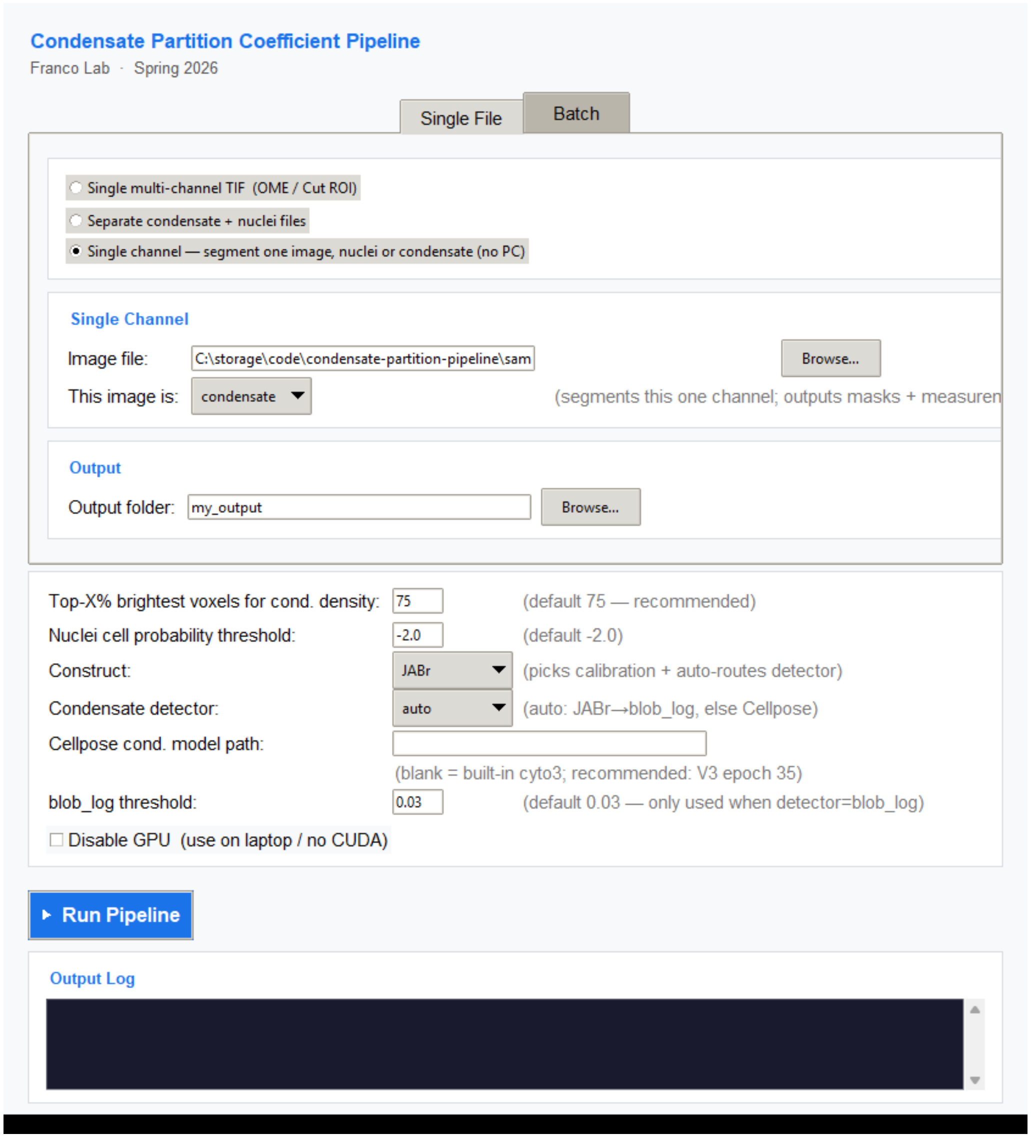
The Single-File tab configured for a JABr run: a multi-channel TIF loaded, Construct = JABr.



After the run: the Output Log reports the calibrated partition coefficient and the status reads Done.

4.2 Single-channel mode

A third input mode, **Single channel**, segments just one image — a condensate *or* a nuclei image — when you do not have both channels. Pick the type from the drop-down; nuclei use Cellpose `cyto3` and condensate uses the same detector the full pipeline would (for `JABr`, `blob_log`, with the intra-nuclear gate disabled so every detection is kept). It outputs masks, object count, volumes, and per-slice measurements — but **no partition coefficient**, which requires both channels.



Single-channel mode: segment one condensate (or nuclei) image on its own.

5 Understanding the settings

For routine JABr analysis you only need to set **Construct**; the rest have sensible defaults.

Setting	Default	What it does
Construct	(none)	Pick **JABr** for the validated workflow. Auto-selects the `blob_log` detector *and* the JABr calibration so the reported PC is on the manual scale.
Condensate detector	auto	`auto` routes JABr→`blob_log`, else Cellpose. Normally left on auto.

Setting	Default	What it does
Top-X% brightest voxels	75	Defines condensed-phase density; trims the dim mask halo to match the manual reference. Leave at 75.
Nuclei cell-prob. threshold	-2.0	How aggressively nuclei are detected. -2.0 suits the lab's nuclear stain.
blob_log threshold	0.03	Condensate spot-detection sensitivity (used when detector = `blob_log`). Leave at 0.03 for JABr.
Cellpose cond. model path	(blank)	Advanced; a fine-tuned condensate model. Leave blank for JABr.
Disable GPU	off	Tick on a laptop with no NVIDIA GPU.

The one rule for routine use: set **Construct = JABr** and run. Everything else is for experimentation or other constructs.

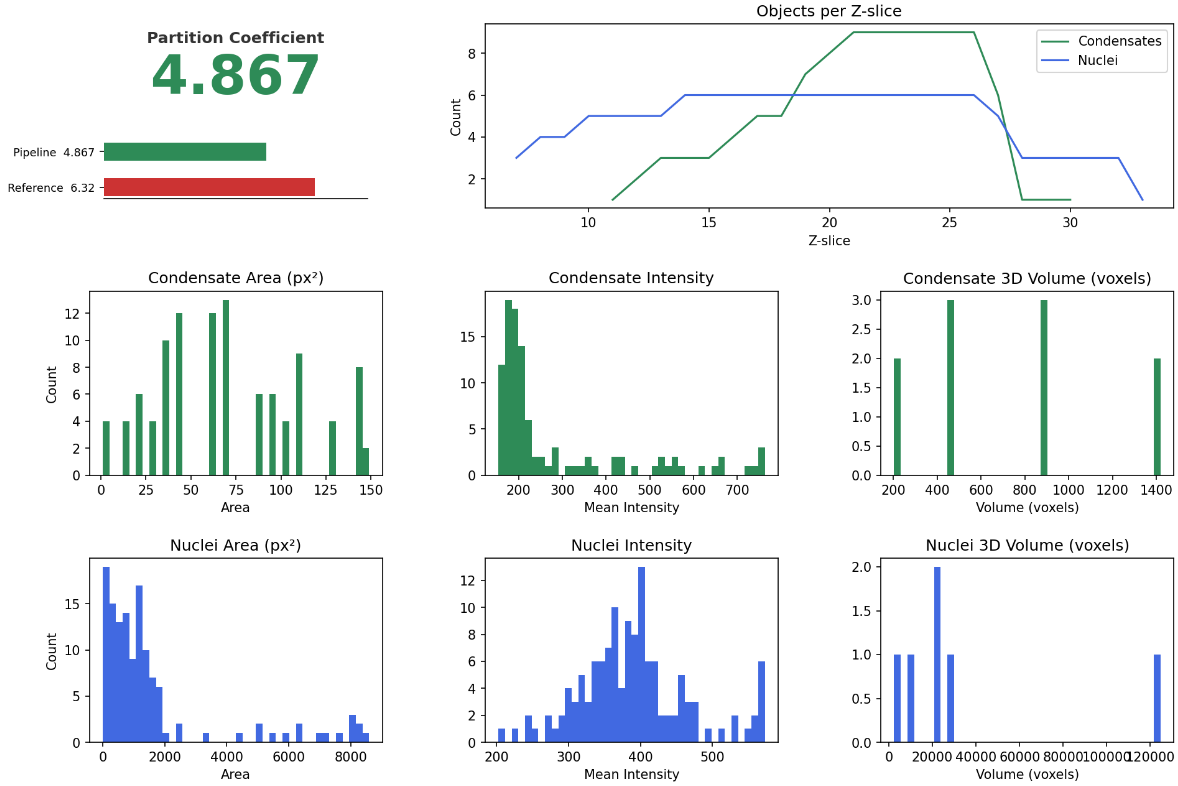
6 Reading the outputs

Every run writes: `summary.csv` (headline numbers incl. the calibrated PC), `results.png` (summary figure), `condensate_masks.tif` and `nuclei_masks.tif` (3D labeled masks), and per-object/per-slice CSVs.

For the bundled sample `JABr_Sample2_5_3.tif`, a real run produces:

`[nuclear] PC raw 4.867` → `calibrated 4.813` (manual reference 4.558, +5.6%)

Report the calibrated nuclear PC (the raw value is systematically ∼3× the manual scale by construction; calibration standardizes it). **Always glance at `condensate_masks.tif`** over the raw channel in Fiji/ImageJ — a 10-second visual check catches the rare misfire.



`results.png` from the bundled-sample run.

7 Batch mode

On the **Batch** tab, select a folder of `.tif` files; every file is processed. Optionally supply the manual Imaris nuclear-PC CSV as a **Reference CSV** — the pipeline then writes `comparison.csv` (pipeline vs manual, per cell) and `scatter.png` with the correlation, RMSE, and MAE. Set **Construct = JABr** and run.

8 Command-line use

```
python pipeline.py --roi sample_data/JABr_Sample2_5_3.tif --construct JABr --output
my_output
```

Separate channels: `--nuc nuclei.tif --cond condensate.tif`. Single-channel segmentation (one image, no PC): `--single condensate.tif --channel condensate` (or `--channel nuclei`). Physical volumes in μm^3: add `--voxel-xy 0.065 --voxel-z 0.3`. Force CPU: `--no-gpu`. For many files (and optional comparison against a manual reference CSV), use the GUI **Batch** tab.

9 Troubleshooting

Symptom	Likely cause / fix
PC wildly off / nuclei look empty	Channels swapped: confirm ch0 = nuclei, ch1 = condensate.
"0 condensates detected"	Not JABr, or very dim signal. Lower `blob_log` threshold (e.g. 0.02); confirm the raw channel has spots.
Very slow (minutes per cell)	Running on CPU. Install a CUDA PyTorch build; ensure **Disable GPU** is unchecked.
`CUDA out of memory`	Tick **Disable GPU**, or close other GPU programs.
Cellpose download fails	First run needs internet for model weights.
One cell's calibrated PC looks off	Calibration is population-level; individual cells vary (±20% for ~80% of cells). Verify against the mask.

10 Method summary

The PC is the ratio of condensed-phase to dilute-phase density (Fabrini et al.), both camera-background-subtracted. Condensed density uses the brightest 75% of voxels inside (condensate ∩ nucleus); dilute density is the mean of the 50 lowest-intensity 10×10×10 patches in the nuclear dilute region. Nuclei are segmented with Cellpose 3 (`cyto3`, 3D), cleaned, and void-filled; condensates are detected with Laplacian-of-Gaussian `blob_log` and kept when ≥ 50% of the rendered sphere lies inside a nucleus. A per-construct linear (JABr) or isotonic calibration standardizes the automated PC onto the manual scale. Full rationale and the cross-construct scope/limits are in `docs/METHODS.md`.

Scope. Validated and production-ready for **JABr**. The detector is construct-specific; a leave-one-construct-out test confirmed a single detector does not generalize zero-shot (it under-detects on unseen constructs). Extending to other constructs needs per-construct calibration or few-shot retraining — see `docs/METHODS.md`, Section 5.